JOB MARKET INTELLIGENCE

Team 1

Team Members

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1. Project Planning & Management

- ## Project Proposal

  Overview
  Job Market Intelligence & Skills Analytics is a multi-source data platform that helps Egyptian and Arab engineers discover job opportunities in Arab countries and abroad, and understand the skills those opportunities require. The system ingests job postings from job-search REST APIs and open datasets, integrates them into one unified schema, extracts the requested skills, and presents the results in an interactive dashboard that compares Arab-market, international and remote opportunities.

- ## Problem statement

  Engineers in Egypt and the Arab region face limited local opportunities. Information about openings in other Arab countries and abroad is scattered across many platforms, in different languages and formats, with duplicated postings and inconsistent job titles. As a result, it is hard to see where opportunities are concentrated, which skills they require, and how local demand compares with international and remote demand. This project consolidates that information into one evidence-based, searchable view.

- ## Aim

  Build a reproducible data pipeline and dashboard that consolidate engineering job postings from multiple sources and reveal where opportunities exist and which skills they ask for.

- ## Objectives

  Build a multi-source ingestion pipeline (REST API collection plus open datasets).

  Integrate all sources into a unified schema with source, country, market type (Arab, International, Remote), language, title, description and salary.

  Clean the data, remove duplicates across sources, and normalize Arabic and English text.

  Analyze opportunity volume by country and market type, skill demand, and the skills gap between Arab and international postings.

  Train and evaluate a baseline classifier that assigns each posting to a role category.

  Deliver an interactive dashboard with filters and an “Opportunity Finder / What Should I Learn?” page.

  Document, test and publish the code and all required documents on GitHub.

- **Data sources**

Source	Type	Coverage	Role in the project
Jooble	Job-search aggregator, REST API	Egypt and Arab countries	Main source of current Arab-market postings
Careerjet	Job-search aggregator, REST API	Egypt and Arab countries	Main source, cross-checked against Jooble
Wuzzuf dataset	Job-board dataset	Egypt	Egyptian market baseline
ArabJobs corpus	Open Arabic job-ad corpus (8,546 postings)	Egypt, Saudi Arabia, UAE, Jordan	Arabic text baseline; about 450 engineering-related postings
GitHub international opportunities dataset	Open dataset	International	International postings
RemoteOK and We Work Remotely	Remote job boards	Global remote	Supplementary remote postings (about 60 collected so far)

Scope

- *Included:* multi-source ingestion, schema integration, cleaning and deduplication, skills extraction, comparative analysis (Arab vs international), baseline ML classification, dashboard, testing and documentation.

- *Out of scope:* CV matching, applying to jobs on behalf of users, redistributing raw third-party data, guaranteeing placement, and conclusions about the entire job market beyond the collected sample.

Expected deliverables

Documented unified dataset and data card, pipeline source code, analysis notebooks, classifier with evaluation report, dashboard, test report, user manual and technical documentation.

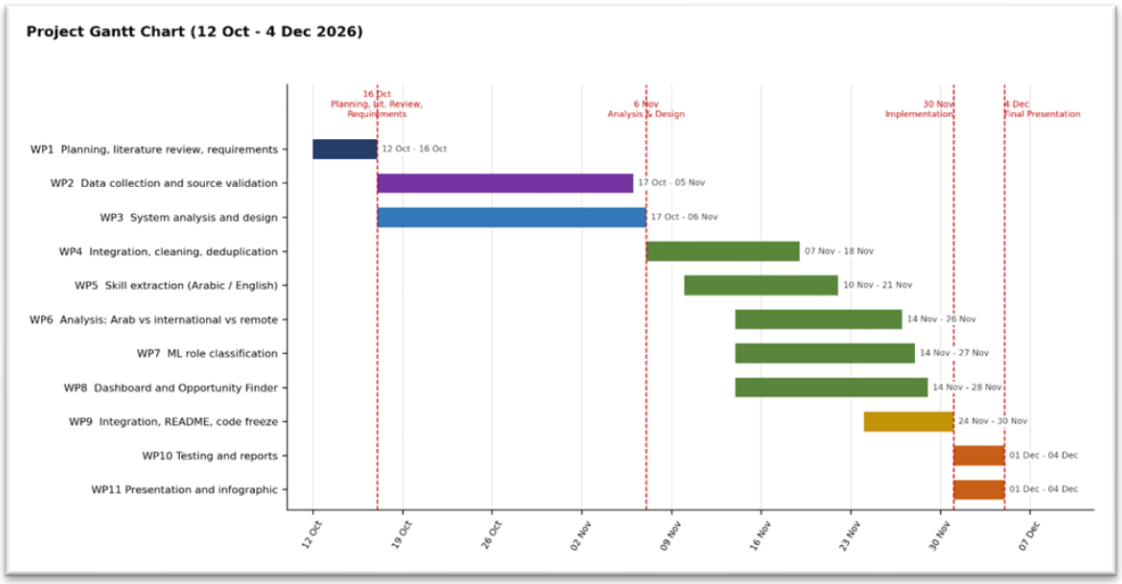
- **Project Plan**
  Work packages and deliverables

WP	Period	Work package	Deliverable
WP1	12 to 16 Oct	Planning, literature review, requirements	Project documentation, Gantt chart, roles, risks, KPIs, GitHub repository
WP2	17 Oct to 5 Nov	Data collection and source validation	Collected postings from all sources, source and license record, data profile
WP3	17 Oct to 6 Nov	System analysis and design	Use case, ER, DFD, sequence, activity, state, class diagrams; architecture; wireframes
WP4	7 to 18 Nov	Integration, cleaning, deduplication	Unified dataset and data card, cleaning scripts
WP5	10 to 21 Nov	Skill extraction (Arabic and English)	Bilingual skill dictionary, extraction module, validation on a 100-posting sample
WP6	14 to 26 Nov	Analysis: Arab vs international vs remote	Analysis notebooks, opportunity and skills-gap findings
WP7	14 to 27 Nov	ML role classification	Trained baseline, evaluation report
WP8	14 to 28 Nov	Dashboard and Opportunity Finder	Streamlit application
WP9	24 to 30 Nov	Integration, README, code freeze	End-to-end system, README
WP10	1 to 4 Dec	Testing and reports	Test plan and report, user manual, technical documentation
WP11	1 to 4 Dec	Presentation and infographic	Final presentation, infographic

Milestones

Date	Milestone	Type
16 Oct	Planning, Literature Review, Requirements submitted	DEPI
5 Nov	Data freeze: no new collection after this date	Internal
6 Nov	System Analysis & Design submitted	DEPI
15 Nov	Minimal end-to-end version (load, clean, one chart)	Internal
30 Nov	Implementation submitted, code freeze	DEPI
4 Dec	Final presentation, testing and reports	DEPI

Gantt chart:



Task Assignment & Roles

Member	Role	Primary responsibilities	Deliverables owned	Backup reviewer
Member 1: [Nada Nasser Osman]	Data Engineer	API collection (Jooble, Careerjet, remote boards), dataset intake (Wuzzuf, ArabJobs, GitHub,hemalays,levretech,googlejobs,we work remotely,remoteok,Kaggle(linked jobs), source and license record, schema integration, cleaning, cross-source deduplication	Unified dataset, data card, data profile	Nada Nasser[Team Leader]
Member 3: [Nada Nasser & Ethar Essam]	Data Analyst	Exploratory analysis, Arab vs international vs remote comparison, skills-gap findings, chart specifications, cross-check of dashboard figures	Analysis notebooks, findings report	Nada Nasser[Team Leader]
Member 4: [Mayar Melad & Abd Elrahman Shawkat]	ML Engineer	Role-category classifier, train/test design, metrics, error analysis	Trained model, evaluation report	Nada Nasser[Team Leader]
Member 5: [Abd ELHameed ALshorbagy]	Dashboard and Integration Lead	Streamlit dashboard, Opportunity Finder page, integration, repository structure, README, testing, demo	Dashboard, README, test plan and report	Nada Nasser[Team Leader]

1.4 Risk Assessment & Mitigation Plan

Likelihood (L) and Impact (I) are rated Low, Medium or High. Risks are reviewed in the weekly team meeting, and the status of each is recorded in GitHub Issues.

ID	RISK	L/I	Mitigation	Contingency
R1	Restrictions on API access or source usage may limit data collection	Medium / High	Check source terms and available access limits before collecting data. Record the sources used and follow their usage conditions.	Use available open datasets and continue with the data already collected.
R2	Some markets may have fewer job postings than others, limiting reliable comparisons.	High / High	Record the number of postings available for each market and assess data coverage before analysis.	Present comparisons as exploratory and clearly state sample-size limitations.
R3	The same job posting may appear in multiple sources, leading to inaccurate counts.	High / Medium	Identify and remove exact duplicates and review likely duplicates using available fields such as title, company, location, and URL.	Report any remaining duplication limitations in the project documentation.
R4	Different data formats and missing fields across sources may complicate integration and analysis.	High / Medium	Use a unified schema and standardize important fields where possible. Check missing values before analysis.	Exclude unreliable fields from specific analyses and document the limitations.
R5	Arabic and English text variations may reduce the accuracy of skill extraction and role classification.	High / Medium	Normalize text, use a manageable bilingual skill dictionary, and evaluate the baseline model using suitable metrics.	Report the limitations of extraction and classification results without making unsupported claims.
R6	Limited time or technical difficulties may delay dashboard integration and testing.	Medium / High	Develop the essential features first and keep the code and documentation organized during development.	Prioritize the core dashboard, simplify optional features, and document any incomplete functionality.

1.5 Key Performance Indicators (KPIs)

Targets are the team's commitments. Those marked with * will be reviewed after the first data profile (31 Oct) and the first model baseline, and any change will be recorded in the project log.

ID	KPI	Target	How it is measured
Data			
D1	Source coverage	Collect and document usable job-posting data from the available sources selected for the project.	Record the sources used and the number of records collected from each source.
D2	Deduplication	Identify and remove confirmed duplicate records where possible during data cleaning.	Check for duplicate records before and after data cleaning.
D3	Required Field Completeness	Check the completeness of key fields, including job title, country, source, and market type, where available.	Data validation summary
ML			
M1	Role classifier*	Weighted F1 of at least 0.70 on a held-out 20% test set; macro-F1 and per-class results reported	Evaluation report
M2	No data leakage	Leakage checklist passed (split before vectorizing, label-source field excluded or tested)	Review by backup reviewer
System			
S1	Dashboard accuracy	10 of 10 sampled dashboard figures match the analysis outputs	Manual cross-check
S2	Response time	Ensure that the system responds to user actions and displays results within a reasonable time.	Timed tests
S3	Availability	Ensure that the system is accessible and can be run successfully in the intended project environment.	Test system accessibility and record whether it runs successfully.
Users			
U1	Usability	Ensure that users can complete the main system tasks easily and without unnecessary assistance.	Observe users while they perform selected tasks and record any difficulties.
U2	User adoption and usefulness	Assess whether users find the system useful for exploring job opportunities and identifying relevant skills.	Collect user feedback through a short survey after trying the system.

2. Literature Review

- **Feedback & Evaluation**
  This section is completed by the lecturer. The team records its response and the action taken after each review.

Date	Milestone reviewed	Lecturer's feedback	Team response / action taken	Status
16 Oct 2026	Planning, Literature Review, Requirements	***: I have reviewed the project documentation, this is an exceptionally well-structured, thorough, and professional project proposal. The scope is well-defined, the methodology is realistic, and the risk management and metrics (KPIs/NFRs) are robust.***	.....	Pending
6 Nov 2026	System Analysis & Design	To be confirmed	To be confirmed	Pending
30 Nov 2026	Implementation	To be confirmed	To be confirmed	Pending
4 Dec 2026	Final Presentation, Testing & Reports	To be confirmed	To be confirmed	Pending

- **Suggested Improvements**

#	Improvement	Benefit	Priority	Planned phase
1	Add more international and remote sources	Stronger Arab vs international comparison	High	WP2, before the 5 Nov data freeze
2	Normalize salaries and currencies across countries	Comparable pay information	Medium	Future work
3	Improve handling of Arabic dialects and mixed Arabic/English text	Better extraction accuracy	Medium	WP5

#	Improvement	Benefit	Priority	Planned phase
4	Personalized matching from a user's skills or CV	More useful recommendations	Low	Future work, after privacy and licensing review
5	Deploy the dashboard online	Easier access for pilot users	Low	If time permits
6	Incorporate lecturer and pilot-user feedback	Closer fit to user needs	High	Continuous

Final Grading Criteria

Weights are not set by the team. They will be filled from the official DEPI assessment form or the instructor's breakdown.

Component	What is assessed	Evidence in this project	Weight
Documentation	**Planning, requirements, analysis and design, technical documentation, user manual**	**The documents in the GitHub repository, delivered on the DEPI milestone dates**	**[to be confirmed]**
Implementation	**Working system, code quality, repository, README**	**Pipeline, dashboard, source code, commit history, README**	**[to be confirmed]**
Testing	**Test plan, test cases, automated tests, bug reports**	**Test report, automated tests, pilot-user testing**	**[to be confirmed]**
Presentation	**Final presentation, infographic, demo, each member's contribution**	**Slides, infographic, demo, per-member responsibilities**	**[to be confirmed]**

Per the DEPI instructions, the final evaluation is based on the final presentation, which must show the delivered outputs and each member's responsibility, and the trainer also gives each trainee a weekly individual evaluation.

3. Requirements Gathering

3.1 Stakeholder Analysis

ID	Stakeholder	Type	Description	Needs and expectations	How the system addresses them	Influence	Interest
SH-01	Egyptian and Arab engineers (job seekers and fresh graduates)	Primary user	Engineers looking for work in Egypt, other Arab countries or abroad	See where opportunities exist; know which skills are requested; compare Arab, international and remote options; results in Arabic and English	Opportunity dashboard, filters by country and market type, “Opportunity Finder / What Should I Learn?” page	Medium	High
SH-02	Engineers targeting international or remote work	Primary user	Subgroup interested in jobs abroad or remote roles	Know which international and remote roles exist and which skills and tools they ask for compared with local roles	Arab vs international vs remote skills-gap view	Medium	High
SH-03	Career advisors, university career offices, training centers	Secondary user	People who guide engineers or design courses	Evidence of market demand to support advice and align training; clear data limitations	Summary charts, data source and limitations page	Medium	Medium
SH-04	DEPI lecturer and evaluators	Reviewer	Assess the project at each milestone	Complete and accurate documents on time; clear division of work; a working, reproducible system; a clear demo	Milestone deliverables on GitHub, README, final presentation	High	High
SH-05	Project team (5 members)	Builder	Design, build, test and present the system	Clear scope, shared schema, fair workload, reproducible	Roles, plan, risk register, repository workflow	High	High

ID	Stakeholder	Type	Description	Needs and expectations	How the system addresses them	Influence	Interest
				environment, access to data			
SH-06	Data providers: Jooble, Careerjet, RemoteOK, We Work Remotely, Wuzzuf and GitHub dataset owners, ArabJobs authors	External provider	Own or publish the job data used	Compliance with terms and licenses; attribution; no redistribution of raw data	Terms record in the data card, source attribution in the dashboard	High	Low
SH-07	Employers and recruiters whose postings are analyzed	Affected party	Authors of the postings	Contact details not exposed; postings not misrepresented	Removal of emails and phone numbers; source links shown	Low	Low

User Stories & Use Cases

Definitions used in this section.

Term	Meaning
Market type	**Arab**: posting located in an Arab country. **International**: posting located outside the Arab region. **Remote**: posting marked remote or collected from a remote job board.
Role category	The engineering role group assigned to a posting (final list to be fixed in the data profile).
Skill	A tool, technology, language or qualification found in a posting by the skill dictionary.

3.2.1 User Stories

Priority follows MoSCoW (Must, Should, Could).

ID	Stakeholder	User story	Acceptance criteria	Priority
US-01	SH-01	As an engineer, I want to see how many opportunities exist per country and market type, so that I know where demand is.	A chart shows posting counts by country and market type; counts match the unified dataset; the sample size is displayed.	Must
US-02	SH-01	As an engineer, I want to filter results by role category, country and market type, so that I see only what is relevant to me.	Each filter updates all views; an empty result shows a clear message.	Must
US-03	SH-01	As an engineer, I want to see the skills most requested for a role category, so that I know what to prepare.	A ranked skills list shows counts and the number of postings analyzed.	Must
US-04	SH-02	As an engineer interested in working abroad or remotely, I want to compare skills requested in Arab, international and remote postings, so that I can see the skills gap.	Side-by-side view of top skills per market type; skills that appear mainly in one market are highlighted; a warning appears when a group has too few postings.	Must
US-05	SH-01, SH-02	As an engineer, I want to choose a target role and market and get the skills to learn first, so that I can plan my learning.	The page lists skills ranked by frequency and explains how they were counted.	Should
US-06	SH-01	As an engineer, I want to browse matching postings with a link to the original source, so that I can apply.	Posting list shows title, country, market type, source and link; the source is attributed.	Should
US-07	SH-01	As a user, I want to paste a job description and see its predicted role category, so that I can understand how the classifier works.	A predicted category with confidence is shown; the evaluation limits are linked.	Could

ID	Stakeholder	User story	Acceptance criteria	Priority
US-08	SH-03	As a career advisor, I want to download a chart or table, so that I can use it in guidance sessions.	Charts export as an image and tables as CSV.	Could
US-10	SH-05	As a team member, I want to run the full pipeline by following the README, so that results are reproducible.	A new member runs the pipeline on a clean environment without help.	Must
US-11	SH-05	As a data engineer, I want each source handled by its own module, so that a source can be added or removed without changing the rest.	Each source has a separate loader that outputs the unified schema.	Should

3.2.2 Actors

Actor	Description
Job Seeker (Engineer)	Primary user of the dashboard
Career Advisor	Secondary user of the dashboard
Team Member (Data Engineer)	Runs and maintains the pipeline

3.2.3 Use cases (summary)

ID	Use case	Primary actor	Main flow	Outcome
UC-01	Explore opportunities	Job Seeker	Open dashboard, choose country, market type and role filters, view charts	Charts update for the selection
UC-02	View requested skills	Job Seeker	Select a role category, open the skills view	Ranked skills with counts and sample size
UC-03	Compare markets (skills gap)	Job Seeker	Select a role, open the comparison view	Top skills for Arab, International and Remote side by side
UC-04	What Should I Learn?	Job Seeker	Choose target role and market, open the page	Skills to learn, ranked, with method explained

ID	Use case	Primary actor	Main flow	Outcome
UC-05	Run the data pipeline	Team Member	Clone the repository, follow the README, run the pipeline	Unified dataset, quality report and outputs produced
UC-6	Collect source data	Team Member, External Data Sources	Call each API or load each dataset, store raw files with source and date	Raw data stored with collection log

3.2.4 Detailed use case descriptions (key use cases)

UC-03: Compare markets (skills gap)

Field	Description
Actor	Job Seeker
Preconditions	The unified dataset and extracted skills are loaded
Main flow	1. User opens the comparison view. 2. User selects a role category. 3. System calculates skill frequencies per market type. 4. System displays the top skills for Arab, International and Remote side by side. 5. System highlights skills that appear mainly in one market.
Alternative flow	If a market type has fewer postings than the minimum threshold, the system shows a “too few postings” warning and hides the ranking for that group.
Postconditions	The user sees which skills differ across markets, with sample sizes

UC-04: What Should I Learn?

Field	Description
Actor	Job Seeker
Preconditions	Skills have been extracted for the selected role and market
Main flow	1. User selects a target role and target market. 2. System ranks the skills by frequency in the matching postings. 3. System shows the ranked list with counts and the number of postings analyzed. 4. System explains how the skills were counted.
Alternative flow	If no postings match, the system suggests broadening the filters.
Postconditions	The user has a prioritized skills list

UC-09: Run the data pipeline

Field	Description
Actor	Team Member
Preconditions	Python environment installed; API keys configured; README followed
Main flow	1. Member runs the collection step. 2. System loads each source with its own module. 3. System maps fields to the unified schema. 4. System cleans text and removes duplicates across sources. 5. System extracts skills. 6. System writes the unified dataset and quality report.
Alternative flow	If an API fails or a quota is reached, the system logs the error and continues with the other sources.
Postconditions	Outputs are ready for analysis and the dashboard

The use case diagram and the remaining detailed descriptions are part of System Analysis & Design (6 Nov).

3.3 Functional Requirements

Priority follows MoSCoW. “Related” refers to the user stories (US) and use cases (UC) in Section

A. Data Collection and Ingestion

ID	Requirement	Priority
FR-01	The system shall collect job postings from the Jooble and Careerjet REST APIs for configured search terms and countries.	Must
FR-02	The system shall load the Wuzzuf, ArabJobs and GitHub international datasets from files.	Must
FR-03	The system shall collect postings from the RemoteOK and We Work Remotely job boards.	Should

B. Integration, Cleaning and Quality

ID	Requirement	Priority
FR-04	The system shall assign each posting a market type (Arab, International, Remote) using a documented rule.	Must
FR-05	The system shall detect the language (Arabic or English) of each posting.	Should
FR-06	The system shall remove emails and phone numbers from posting text before storage or display.	Must
FR-07	The system shall normalize Arabic text (diacritics, letter variants, punctuation) and standardize English text.	Must
FR-08	The system shall standardize country names and job titles.	Should
FR-09	The system shall remove exact and near-duplicate postings across sources, keeping the list of sources for each kept posting.	Must
FR-10	The system shall keep only engineering postings, using documented role rules.	Must
FR-11	The system shall produce a data-quality report per source and for the unified dataset (record counts, missing values, duplicates, description length, language mix).	Must

C. Skill Extraction

ID	Requirement	Priority
FR-12	The system shall extract skills from the title and description of each posting.	Must
FR-13	The system shall store the extracted skills linked to each posting.	Must

D. Analysis

ID	Requirement	Priority
FR-14	The system shall count postings by country, market type, role category and source.	Must
FR-15	The system shall calculate skill frequencies by role category, country and market type.	Must
FR-16	The system shall calculate the skills gap between market types and flag groups below a minimum number of postings.	Must
FR-17	The system shall summarize salaries where available, showing coverage and currency.	Could

E. Machine Learning Classification

ID	Requirement	Priority
FR-18	The system shall train a classifier that predicts the role category from posting text, using a stratified train/test split with a fixed seed.	Should

ID	Requirement	Priority
FR-19	The system shall report weighted F1, macro-F1 and per-class results.	Should
FR-20	The system shall predict the role category of a pasted job description.	Could

F. Dashboard

ID	Requirement	Priority
FR-21	The dashboard shall provide filters for country, market type, role category and source.	Must
FR-22	The dashboard shall show an opportunities overview by country and market type.	Must
FR-23	The dashboard shall show the most requested skills for the selected filters.	Must
FR-24	The dashboard shall provide a market comparison view (Arab, International, Remote).	Must
FR-25	The dashboard shall include a “What Should I Learn?” page.	Should

G. Operations and Documentation

ID	Requirement	Priority
FR-26	The pipeline shall run end to end from a clean clone by following the README.	Must
FR-27	The repository shall include a README with installation steps, system requirements, configuration, and an execution guide.	Must
FR-28	The pipeline shall log record counts at each stage.	Should

3.4 Non-functional Requirements

Targets are the team’s commitments and will be measured as described. Items marked * depend on the first measurements and may be revised and recorded in the project log.

ID	Category	Requirement	Measure / acceptance	Related KPI
Performance				

ID	Category	Requirement	Measure / acceptance	Related KPI
NFR-01	Performance	The dashboard shall load and respond quickly on a standard laptop.	Initial load under 5 seconds; filter updates under 3 seconds.	S2
NFR-02	Performance	The processing pipeline (excluding API collection) shall complete in a reasonable time.*	Full run on the unified dataset under 30 minutes; time recorded per stage.	P1
NFR-03	Scalability	The system should process the available dataset efficiently. .	The system is tested using the available dataset, and any performance limitations are documented.	D2
Security and Privacy				
NFR-04	Security	API keys and credentials shall never be stored in the repository.	Keys loaded from environment variables or an untracked config file; repository scan finds none.	P3
NFR-05	Privacy	Personal data (emails, phone numbers) shall not be stored or displayed.	Automated scan of the final dataset finds none.	D4
NFR-06	Security	User input (pasted job descriptions) shall be validated.	Length limit and empty-input handling; no input reaches the file system or shell.	S4
NFR-07	Security	Project dependencies shall be documented and limited to necessary libraries.	`requirements.txt` with versions; no unused packages.	P1
NFR-08	Compliance	Data shall be used only in line with each source's terms and license.	Terms and license recorded per source in the data card; attribution shown in the dashboard; raw third-party data not redistributed.	D1
Usability				
NFR-09	Usability	A first-time user shall complete the main tasks without help.	5 pilot users, 4 tasks, at least 80% success.	U1
NFR-10	Usability	Arabic text shall display correctly, including right-to-left text.	Visual check of Arabic labels, tables and sample postings.	U1
NFR-11	Usability	Every chart shall be self-explanatory.	Each chart has a title, labelled axes, and the sample size.	U1, S1

ID	Category	Requirement	Measure / acceptance	Related KPI
NFR-12	Accessibility	Charts shall not rely on color alone and shall use readable contrast.	Review of colors, labels and text sizes.	U1
NFR-13	Usability	The interface shall use clear and consistent labels in the selected interface language.	Interface labels and displayed content are reviewed for clarity and consistency.	U2
Reliability				
NFR-14	Reliability	The pipeline shall handle API failures and quota limits.	Up to 3 retries with waiting time, then log and continue with other sources.	P1
NFR-15	Reliability	The dashboard shall not crash on empty or unusual selections.	Tests for empty filters, missing fields and very small groups.	S4
NFR-16	Reliability	The system should provide clear instructions for accessing and running the application in the available environment.	The application can be launched by following the documented instructions in the available environment.	S3
NFR-17	Reliability	Original collected data should be preserved before cleaning and transformation.	Original data files are retained separately from processed datasets.	D1
Data Quality and Transparency				
NFR-18	Data quality	Missing values in important fields should be identified and their potential impact on analysis should be considered.	Missing values in key fields are examined, and relevant limitations are documented.	D4
NFR-19	Data quality	Duplicates shall be removed across sources.	Zero exact duplicates in the final dataset.	D3
NFR-20	Transparency	The project should document the available data sources, dataset size, and relevant data limitations.	Data sources and record counts are documented where available, and known data limitations are identified.	S1
NFR-21	Accuracy	The quality of skill extraction and any implemented machine learning model should	Appropriate evaluation methods are used for implemented components,	N1, M1

ID	Category	Requirement	Measure / acceptance	Related KPI
		be evaluated where applicable.	and their results and limitations are documented.	
Maintainability, Portability and Reproducibility				
NFR-22	Maintainability	Code shall be modular, readable and documented.	One module per source; docstrings on public functions; consistent style; no single oversized script.	P3
NFR-23	Maintainability	Changes shall be reviewed.	100% of pull requests reviewed by the backup reviewer.	P3
NFR-24	Testability	Core logic shall be covered by automated tests.	At least 15 passing tests for cleaning, normalization and skill extraction.	S4
NFR-25	Portability	The project shall run on common systems.	Runs on Windows and one other system (macOS or Linux) from the README.	P1
NFR-26	Reproducibility	Results shall be repeatable.	Fixed random seeds; environment documented; 3 of 3 clean-clone runs succeed.	P1
NFR-27	Documentation	Documentation shall be complete and delivered on time.	All DEPI milestones on GitHub by their deadlines.	P2